Article Critical Review

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Geography 5301

"Free-grazing Duck and Highly Pathogenic Avian Influenza, Thailand"

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Introduction

Insdfsdf 2004 and 2005, Thailand experienced an outbreak of Highly Pathogenic Avian Influenza (HPAI) creating three separate epidemics. This outbreak is a result of the spread of the H5N1 virus. This caused many deaths in Thailand and posed a threat throughout the region and the entire world, so there was a need to understand the risk factors associated with the spread of this virus. The research question is thus: What are the risk factors associated with the spread of the H5N1 virus during this second epidemic? This article goes in depth about a study that analyzed the HPAI spatial distribution based on laboratory-confirmed H5N1 outbreaks recorded during the second epidemic (July 2004 – May 2005). Marius Gilbert, the primary author of this article, is a postdoctoral fellow at the Laboratory of Biological Control and Spatial Ecology at the Université Libre de Bruxelles. Many factors were considered as possible reasons for the spread of this virus and the spatial interaction of these factors was the focus of the study. This study analyzed the HPA1 outbreaks in relation to poultry, land use, and other factors involving humans. This is an important study because other countries in Southeast Asia such as Vietnam and Cambodia have a very similar geography where free grazing ducks, wetlands, other animals and humans meet. What is learned here can be used to mitigate the problem in regions of similar geography to Thailand. To identify these risk factors, an autologistic multiple regression was used by the researchers to relate HPAI to the geographic distribution of the main poultry species, relevant land-use features, and to other environmental or anthropogenic variables. The study shows the strongest association between the H5N1 virus with free-grazing ducks in Thailand with other associations with native chickens, cocks, wetlands and humans. The critical factor in the spread of H5N1 is the wetlands where double-crop rice production takes place and free-grazing duck feed year round.

Analysis

The data on the spread of HPAI through the H5N1 virus was conducted from July 3, 2004 to May 3, 2005 by the Department of Livestock Development, Ministry of Agriculture and Cooperatives, Bangkok, Thailand. The number of samples in this study is 1,716 laboratory-confirmed cases in this time period. The data was organized and categorized as in the presence or absence of the virus within the 8,089 sub districts of Thailand for the national level and for 913 villages in the Suphanburi Province at the local level.

It is important to note the "X-ray Survey" conducted by the Thai government starting October 2004 till early November 2004. This survey produced a composite picture of the H5N1 situation in their country. This involved hundreds of inspectors searching door to door for evidence of HPAI. This survey involved recording incidences along with destroying affected poultry. This study produced poultry census data collected simultaneously from all Thai villages countrywide during the October 2004 – November 2004 time frame. Data from the X-ray Survey were gathered by sub district for analysis at the national level or by village for analysis in the Suphanburi Province. This coincided very well with the analysis done in this study, but it was not clear from the article to what extent what data in the survey was actually from the X-ray Survey or not.

The categories in the analysis were farm chickens, native chickens, farm ducks, and free grazing ducks. Free grazing ducks are domestic ducks raised in the open in flocks of >1000 birds for egg production and to a lesser extent, for meat. Free grazing ducks and native chickens are more exposed to the open and are infected by pathogens in a similar way, so they are in a separate category than the other birds in the study. In addition to poultry data, the researchers gathered

relevant variables describing road network, land use, and the physical environment (2). All the variables in the study were categorized in two ways. The first is to study the rice fields, wetlands, and topography to link the introduction and propagation through waterfowl's frequenting rice paddy fields and wetlands to the spread of the virus. The second is to study the variables on the human population and roads to get an idea on the role of human activities has on the spread of the virus.

The preliminary results of the study show that nearly 50% of all the outbreaks in the country occurred in the Suphanburi Province. This province happened to have a high number of free-grazing ducks. This prompted the researchers to perform a presence or absence data analysis on village-level for this province. Stepwise multiple logistic regression was used to study the association between HPAI occurrences, either at the sub district or village level, with the poultry and environmental variables. Efforts were made to minimize data being skewed with spatial autocorrelation by applying an autologistic approach (4, 5) that averages the probability of HPAI presence among a set of neighbors, defined by the limit of autocorrelation weighted by the inverse of the Euclidean distance. This helped ensure the independence of the samples replicated through space (3). Variables were then selected by testing the status of the virus separately against each of the variables including the autocorrelation adjustment. After that, a stepwise multiple logistic regression was used to get the values needed for the analysis. This was repeated for each variable.

Summary of Findings

Most of the cases of HPAI were found in chickens. However, the distribution of the spread of the virus had the strongest association with the distribution of free-grazing ducks (1). In fact, the

spatial distribution of the outbreak of HPAI does not correspond to areas with high densities of chickens. The results are shown in Table 2 and Table 3 of the article. These tables show that the number of free-grazing ducks as the strongest risk factor associated with the presence of HPAI. To a lesser extent, HPAI was also associated with native chickens, land elevation, number of cocks, and size of the human population. Elevation is the only variable showing a negative association with HPAI indicating a positive association with the lower plains. The analysis carried out on the village level for the Suphanburi Province was very consistent with those obtained on the national level. The study shows that the distribution pattern of the virus suggests an important role of free-grazing ducks in rice paddies as in the central plains of Thailand. Ducks have shown to be less susceptible to the virus than other birds therefore making the virus more difficult to notice and contributing to the spread of the disease. In general, it was found that free-grazing ducks form a HPAI risk factor both in chickens and ducks implying that they form a reservoir of HPAI virus. So, free-grazing ducks are more associated with the transmission of the disease and the actual density of chicken or duck population is associated with the probability of the occurrence of the infection. Infections are transmitted in various ways through livestock or poultry either locally through adjacent production units (i.e. wind, insects, or rodents) or over long distances by movements of animals, persons, or infected material (6). Local spread creates clustering of cases as opposed to long distant spread producing cases that are distance independent. The results of the study show a weak autocorrelation of HPAI presence or absence indicating weak clustering therefore supporting the argument that there was a long distance movement of the infection. Free-grazing ducks are associated with double-crop areas of rice paddy fields (shown in figure 4 of the article). These double crop areas are a meeting point of both domestic and wild animals. This may have served as the entry point

of the virus. Migratory birds may have a role in the spread of the virus in wide areas, but the free grazing ducks are still primarily responsible for the spread of the disease once the virus arrives. This role of migratory birds is not known and further studies are needed.

With a strong association between ducks and rice crops, remote sensing can be used to identify rice crop areas where the ducks reside and can be used to document the reservoirs where the infections can take place. In 2005, several control measures were implemented to help alleviate the spread of the H5N1 virus. Many of these control measures targeted free-grazing duck husbandry. This involved registration and surveillance of all flocks, pre-movement testing, and incentives to move away from free-grazing duck husbandry. As a result, only 64 outbreaks were recorded in the same period in 2005. This shows a drastic decrease in the circulation of the virus.

Conclusion

The question posed by the article "What are the risk factors associated with the spread of the H5N1 virus during this second epidemic?" was not only answered, but one risk factor in particular showed to have the highest association with the spread of the virus. This article conclusively shows that free grazing ducks were a critical component in the spread of the H5N1 virus in the second epidemic in Thailand in 2004. Before this epidemic, little was known about how this kind of virus spread in this kind of geography, but now many ways to alleviate the spread of this disease can be implemented and an unnecessary waste of time and resources can be avoided the next time this kind of epidemic happens. This article was very good in how it presented the problem and how it objectively came to its conclusions by effectively analyzing many factors involved with this outbreak and implementing proven statistical techniques to come

to its conclusions. I learned a lot from this article for several reasons. Factually, I learned a lot about the dynamics of the geography of the region and how knowledge of the interaction of nature and disease is important. An even more important lesson learned from this study is that you can't always look at the highest frequency of an occurrence of a variable as in this case of the number of chickens affected by the H5N1 virus and conclude that chickens are the main reason for the spread of the disease. This study objectively looked at all the factors and found that indeed this was not the case. The article was easy to read in many places, but was quite technical of other areas, especially the statistical analysis section. The parts were separated enough to where you can easily get the meaning of the article without reading all the technical sections.

This article builds a strong argument to answer the research question by listing many of the known variables thought to be associated with spread of the H5N1 virus and analyzing each one. The statistical analysis of each variable was explained quite well using autologistic multiple regression to relate the HPAI to many of the variables explained in the analysis. The reason for this analysis were explained quite well along with explaining how to alleviate erroneous data by paying close attention to factors such as spatial autocorrelation. The sample size of was quite large and the results were compared on two different scales of local village such as the Suphanburi Province with that of the whole nation. This adds a sense of robustness to the conclusions. This analysis would even be stronger if this same study is done in other locations of similar geography and/or compiling data in this same area over several years. Of course, this is subject to frequency of the virus throughout the years.

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